Is there any difference in temperatures in Sydney by looking at maximum weekly temperatures 1973 vs 2003

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1 Introduction

This dataset is taken from the Australian government website `www.bom.gov.au` and provides weekly maximum recorded temperatures (in degrees Celsius) at Sydney Airport for the years 1973 and 2003. After examining various statistical summaries, we were motivated to investigate whether there is a statistically significant difference in temperature distributions between these two years. Such a difference could reflect broader climatic changes or local environmental factors over the three-decade span.

Beyond hypothesis testing, we also explore underlying temperature trends using non-parametric smoothing techniques. To address potential boundary biases often observed with kernel-based methods, we consider local polynomial regression (LOESS) as an alternative approach and what these methods can reveal . Overall, this report aims to uncover insights into the temperature dynamics present in this dataset, potentially hinting at larger-scale shifts in climatology and environmental conditions.

2 Exploratory Data Analysis

Here, we aim to motivate a general hypothesis about these datasets from 1973 and 2003 and to do so, we would like to see how the data changes holistically between the years and see what that could imply about the distributions of temperatures.

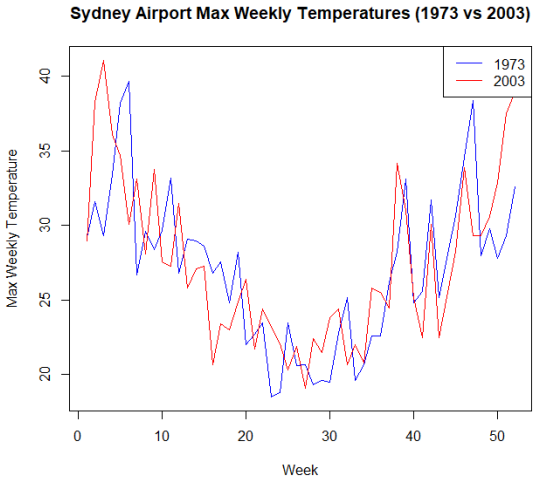


Figure 1: Stochastic time-series plot of the maximum temperatures over the weeks, overlaid together for both years

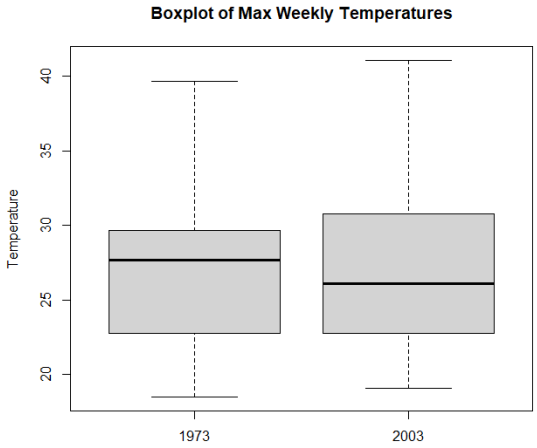


Figure 2: Box-plot for both years with (Q1,Q2,Q3,IQR)

Figure 1 shows that the general trend for both follows the same shape as we would expect since it's the change of temperature throughout the year. However, note also that there are local regions where the red line

is constantly below the blue and constantly above and this switches out. This followed by the observation that the blue line's year-to-year trough-peak jumps seem to be more drastic than the red line's. Overall, this suggests these two samples may come from the same family of distributions

Figure 2 gives the quantiles and interquartile-range (IQR) of $(22.77, 27.70, 29.62, 6.85)$ and $(22.88, 26.10, 30.70, 7.82)$ in degrees Celsius for the years 1973 and 2003 respectively. One might be lead to expect that, for short time periods relative to times of geological activities of Earth, the median should be the same and the spread should also be the same. However, this box-plot shows clear differences in central tendencies (spread) as the median of the of the 2003 temperatures is lower by 1.6. Moreover the IQR is higher and the higher max in 2003 is emblematic of the higher volatility noted in Figure 2. These main observations support the idea that there is a difference in the underlying temperature distributions between the years and guides our motivation for a hypothesis.

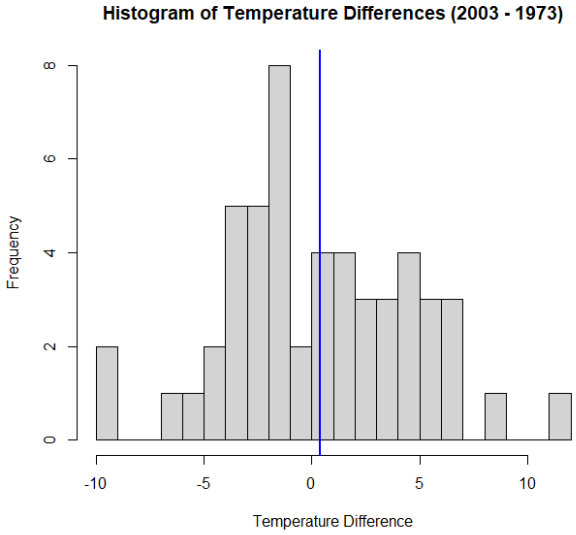


Figure 3: Histogram of the temperature differences between 2003 and 1973 with the blue line noting the mean temperature difference

Figure 3, looks at the distribution of the differences in temperatures every week - the two readings form paired observations as they are readings of the same location and so using this metric can be useful for our hypothesis test. The main things to note is that the mean temperature difference of 0.36 which motivates our belief that there might be a difference in the temperature distribution between the two years. While the mean might suggest a positive difference, the disagreement with the median from Figure 2 means that we can't have a stronger claim about whether this difference is positive or negative; the distribution of the means seems to be mostly centred around zero. Each plot contributes to building a case for testing the hypothesis whether or not there has been a difference in the maximum weekly temperatures between 1973 and 2003.

3 Formulating and testing the hypothesis

3.1 Hypothesis

More formally, the hypothesis we wish to test is:

- Null Hypothesis H_0 : there is no difference in the maximum weekly temperatures between 1973 and 2003 (i.e. the differences are centred around zero)
- Alternate Hypothesis H_1 : there is a significant difference in the maximum weekly temperatures between the two years (i.e the distribution of differences is shifted away from zero)

3.2 Choosing and performing the test

Initially the simplest test we can choose to perform is a t-test. However, we will not perform a t-test here because the normality assumption is heavily questionable. Instead, we will use the Wilcoxon signed-rank test as it's a non-parametric test that relies on the ranks of differences rather than their actual values making it more robust to deviations from normality in the presence of extreme values which could be the case here either due to special weather events or even errors in measurements.

We will use the Wilcoxon signed-rank test (WSRT). In order to use this test we have to make some assumptions and fulfil them: the data are paired (true as they are temperature measurements of the same location at different times), there is symmetry about the median (which is also the mean under the null hypothesis as a result), continuous distribution of differences such that no two values are the same so we get unique ranks and independence of pairs. Whilst we will be using the WSRT, we must also note major caveats. Initially, the independence of pairs: there is not true independence as temperature doesn't randomly change but gradually as it is tied to the movement of the Earth however if the week-to-week correlation is weak then it's a justifiable assumption for simplicity. A much greater concern is the assumption of symmetry about the median (or mean under the null hypothesis); Figure 3 clearly showed that there is some skew and a lack of symmetry about the chosen null of zero which may pose a problem in our test.

Performing the WRST in R, with continuity correction, yields a p-value of 0.6132 (see Appendix for Code). This means for a two-tailed test with significant level of 5 percent, there is insufficient evidence to reject H_0 and so cannot state that there is a non-zero difference in the maximum weekly temperatures between 1973 and 2003.

4 Smoothing with a kernel smoother

We are going to now smooth both the 1973 and 2003 columns using the Nadaraya-Watson (NW) kernel smoother which showcases the kernel-weighted average form - intuitive approach for local averaging. Given observations (x_i, Y_i) with x_i denoting the ith week and Y_i the corresponding observed temperature then the NW estimator of $m(x)$ (the theoretical true smoothed function) at x_i is:

$$\hat{m}(x_i) = \sum_{j=1}^{n} \frac{K(\frac{x_i - x_j}{h})}{\sum_{k=1}^{n} K(\frac{x_i - x_k}{h})} Y_j$$

where $K(\cdot)$ is a kernel function and $h > 0$ is the bandwidth controlling the smoothness of the fit. Here we choose K to be the Gaussian (normal) kernel for its smoothness and well-understood properties

$$K(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right)$$

4.1 Bandwidth selection

The bandwidth h determines the smoothness of our overall smoothed function. To determine the optimal bandwidth we use the leave-one-out cross-validation (LOOCV) in order to balance the bias-variance trade-off by minimising the mean summed square error of the fit (**MSSE(h)**), by omitting each (x_i, Y_i) and evaluating the predictive error. We minimise this with respect to h and choose the appropriate h.
So now that we have completed the LOOCV, we can look at the graphs and choose appropriate bandwidths. Whilst from the graph values such as $h_{1973} = 1$ and $h_{2003} = 2$ minimise the quantity we are looking for, we must remember the bias-variance trade off; if we pick such small bandwidths then we would be fitting the noise as larger bandwidths can reduce sensitivity to outliers and noise which lead to more stable and simpler models. Therefore, we should pick other values for the bandwidth such as points near the elbow-point as the increase in the LOOCV error is miniscule relative to the improved smoothness of our smoothed function. Hence, we will use $h_{1973} = 8$ and $h_{2003} = 8$ for the same reason: the increase in the LOOCV error (relative to the Data Scale) for both is between 1 to 2.5 which is emblematic of a root MSE of about 1 to 1.6 degrees Celsius which is good for the improved smoothness that we will get.

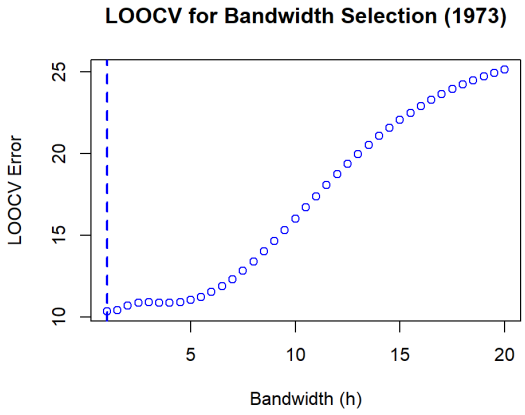


Figure 4: The LOOCV error for the different possible bandwidths for 1973 temperatures

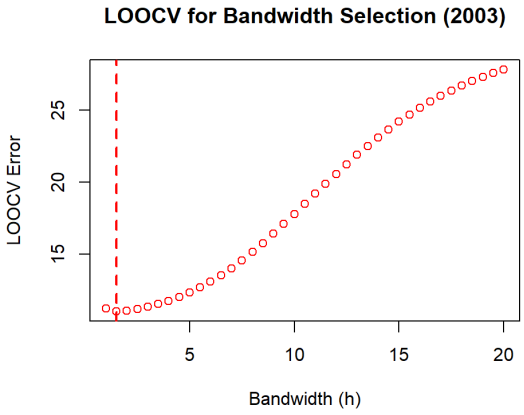


Figure 5: The LOOCV error for the different possible bandwidths for 2003 temperatures

4.2 Smoothed final result

After choosing the appropriate bandwidth, all that is left is to compute the respective smoothed functions. The results produce (see Appendix for exact code):

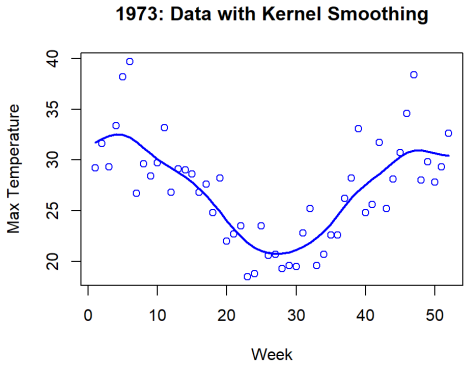


Figure 6: The kernel smoothed out function for 1973 data

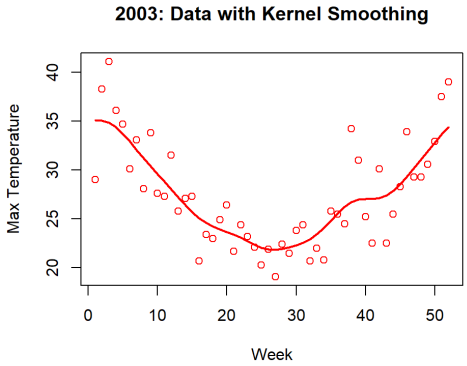


Figure 7: The kernel smoothed out function for 2003 data

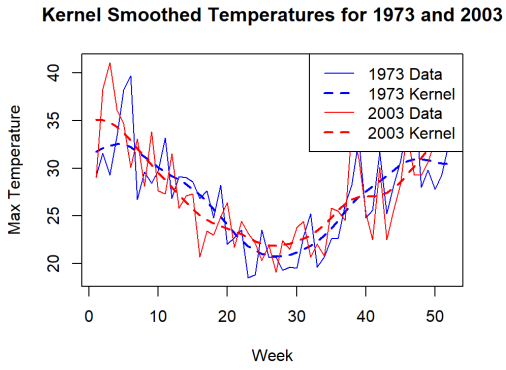


Figure 8: Both kernel smoother temperatures on the same graph

Figure 6 and 7 show NW-smoothed curves for the 1973 and 2003 maximum weekly temperatures. The chosen bandwidth resulted in a visually interpretable smooth function which highlights seasonal patterns (peaks and troughs) while mitigating short-term fluctuations and noise. Figure 8 shows that temperatures in 'wintery' weeks of 2003 (approximately weeks 20 to 39 as Australia is far in the Southern Hemisphere) are colder than they were in 1973 but also the 'summery' weeks of 2003 (the weeks outside of weeks 20 to 39) are hotter than they were in 1973. Whilst previously from our hypothesis test we concluded that there wasn't any sufficient evidence to state that the maximum weekly temperatures differed from the two years, that used the measure of mean which wouldn't show any indication of this behaviour as the increased hot temperatures and decreased cold temperatures could've cancelled out to give the same or similar mean. However, having used NW-smoothers, it illustrated a darker truth: while the average temperatures between the two years doesn't seem to be any significantly different, 2003 has colder cold weather and hotter hot weather which is an indication in the temperatures becoming more extreme which could signify affects of climate change and the climate in Sydney is becoming more extreme in both ends. Hence, looking back at our test, these results might align : in order to correctly perform the test and achieve symmetry, we must've considered a test or several tests on symmetric subsections of the data and thus the results of those tests would have coincided with this observed behaviour of the more extreme temperatures 30 years later.

5 Alternative smoothing procedure

While the NW kernel estimator provides a straightforward approach to smoothing, it can be susceptible to boundary bias and may overemphasise localised fluctuations when the bandwidth is small. This occurs because when we fit a kernel at the boundary data (for example a normal kernel) there isn't an equal influence from both sides as the points on the boundary have more points towards one side of them. To address these valid concerns, we can apply a local polynomial smoothing method known as LOESS. In this procedure, a polynomial (usually degree 1 or 2) is fit within moving windows across the domain whilst observations are weighted by a distance-based kernel. This helps capture local linear trends and often reduces the boundary bias compared to the NW kernel method.

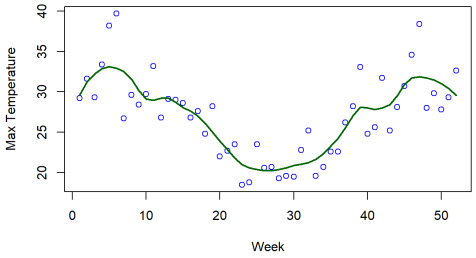


Figure 9: 1973 with local polynomial smoothing

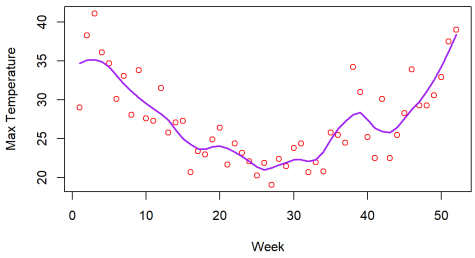


Figure 10: 2003 with local polynomial smoothing

For the LOESS, the span parameter that was chosen was 0.3. The span parameter controls the fraction of the data is used in each local polynomial fit meaning a span of 0.3 means that each local regression is fit to roughly 30 percent of the data points nearest to the point being estimated (in the x space). Once again, there is a bias-variance trade off here; a smaller span is more flexible allowing for fitted curve to adapt to local changed in the data (less bias and more variance) while larger span yields smoother curve that might miss fine details (more bias,less variance). Here 0.3 was chosen in order to yield less bias and in order to pry more into the weekly fluctuations as the NW smoother seemed to have removed those fluctuations. From Figure 11 and Figure 12, we can clearly see an improvement on the sensitivity of the local boundaries and the general sensitivity without compromising the overall trend and shape. The LOESS seemed to capture more fluctuations seasonally which might be of interest. One might prefer the LOESS due to it being more expressive at the boundaries which we are equally interested in as the middle-bulk of the data.

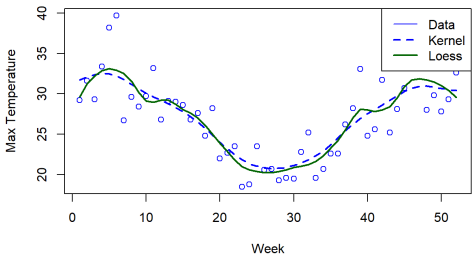


Figure 11: 1973 NW kernel smoothing and LOESS on the same graph

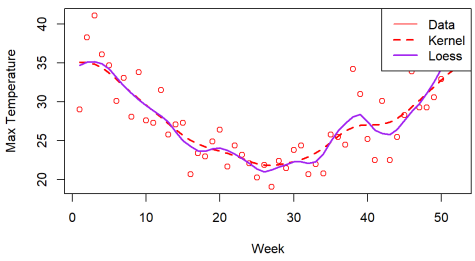


Figure 12: 2003 NW kernel smoothing and LOESS on the same graph

Appendix

```
## Third Stats Practical
library(ggplot2)
library(dplyr)
library(KernSmooth)
### 1. Exploratory Data Analysis
sydtemp <- read.csv(file.choose(), header = TRUE)
# Examine the structure and first few rows
str(sydtemp)
head(sydtemp)lowers
# Summary statistics for the two temperature columns
summary(sydtemp[, c("Temp1973", "Temp2003")])
# Plot the time series for both years on the same graph
plot(sydtemp$Week, sydtemp$Temp1973, type = "l", col = "blue",
     ylim = range(c(sydtemp$Temp1973, sydtemp$Temp2003)),
     xlab = "Week", ylab = "Max Weekly Temperature",
     main = "Sydney Airport Max Weekly Temperatures (1973 vs 2003)")
lines(sydtemp$Week, sydtemp$Temp2003, col = "red")
legend("topright", legend = c("1973", "2003"), col = c("blue", "red"), lty = 1)
# Boxplots to compare the distributions
boxplot(sydtemp$Temp1973, sydtemp$Temp2003,
        names = c("1973", "2003"),
        main = "Boxplot of Max Weekly Temperatures",
        ylab = "Temperature")
# Since the data are collected weekly for the same location, we can form paired
    observations.
# Create a vector of the differences (2003 - 1973)
temp_diff <- sydtemp$Temp2003 - sydtemp$Temp1973
# Histogram of differences
hist(temp_diff, breaks = 20,
     main = "Histogram of Temperature Differences (2003 - 1973)",
     xlab = "Temperature Difference")
abline(v = mean(temp_diff), col = "blue", lwd = 2)
### 2. Hypothesis Formulation and Testing
wilcox_result <- wilcox.test(sydtemp$Temp2003, sydtemp$Temp1973, paired = TRUE)
print(wilcox_result)
### 3. Smoothing using a Kernel (Nadaraya-Watson)
y2003 <- data$Temp2003
h_candidates <- seq(1, 20, by = 0.5)
# Compute LOOCV error for each candidate h for the 2003 data
cv_errors_2003 <- sapply(h_candidates, loocv_error, x = x, y = y2003)
# Identify the optimal bandwidth for 2003
optimal_h_2003 <- h_candidates[which.min(cv_errors_2003)]
cat("Optimal bandwidth for 2003:", optimal_h_2003, "\n")
```

```r
# Plot the LOOCV error versus bandwidth for the 2003 series
plot(h_candidates, cv_errors_2003, type = "b", col = "red",
     xlab = "Bandwidth (h)", ylab = "LOOCV Error",
     main = "LOOCV for Bandwidth Selection (2003)")
abline(v = optimal_h_2003, col = "red", lwd = 2, lty = 2)
x <- data$Week
y1973 <- data$Temp1973
y2003 <- data$Temp2003
# Choose a bandwidth h. (You may experiment with different values or use cross-
    validation.)
h <- 8  # Example value;
# Apply the Nadaraya-Watson kernel smoother (with a normal kernel) for each series.
ksmooth_1973 <- ksmooth(x, y1973, kernel = "normal", bandwidth = h, x.points = x)
ksmooth_2003 <- ksmooth(x, y2003, kernel = "normal", bandwidth = h, x.points = x)
# Plot the original data with the kernel-smoothed curves for each year.
plot(x, y1973, type = "p", col = "blue", main = "1973: Data with Kernel Smoothing",
     xlab = "Week", ylab = "Max Temperature")
lines(ksmooth_1973, col = "blue", lwd = 2)
plot(x, y2003, type = "p", col = "red", main = "2003: Data with Kernel Smoothing",
     xlab = "Week", ylab = "Max Temperature")
lines(ksmooth_2003, col = "red", lwd = 2)
plot(x, y1973, type = "l", col = "blue",
     ylim = range(c(y1973, y2003, ksmooth_1973$y, ksmooth_2003$y)),
     xlab = "Week", ylab = "Max Temperature",
     main = "Kernel Smoothed Temperatures for 1973 and 2003")
lines(x, ksmooth_1973$y, col = "blue", lwd = 2, lty = 2)
lines(x, y2003, col = "red")
lines(x, ksmooth_2003$y, col = "red", lwd = 2, lty = 2)
legend("topright",
       legend = c("1973 Data", "1973 Kernel", "2003 Data", "2003 Kernel"),
       col = c("blue", "blue", "red", "red"), lty = c(1, 2, 1, 2), lwd = c(1, 2, 1,
           2))
### 4. Alternative Smoothing Method: Local Polynomial (loess)
s<-0.3
loess_fit_1973 <- loess(y1973 ~ x, span = s)  # Adjust span as needed (0.2 is an
    example)
loess_fit_2003 <- loess(y2003 ~ x, span = s)
# Predict smoothed values using the loess models.
loess_pred_1973 <- predict(loess_fit_1973, newdata = data.frame(x = x))
loess_pred_2003 <- predict(loess_fit_2003, newdata = data.frame(x = x))
# For 1973:
plot(x, y1973, type = "p", col = "blue", xlab = "Week", ylab = "Max Temperature")
lines(x, loess_pred_1973, col = "darkgreen", lwd = 2)
# For 2003:
plot(x, y2003, type = "p", col = "red",xlab = "Week", ylab = "Max Temperature")
lines(x, loess_pred_2003, col = "purple", lwd = 2)
# 1973: Kernel vs. Loess
plot(x, y1973, type = "p", col = "blue",
     xlab = "Week", ylab = "Max Temperature")
lines(x, ksmooth_1973$y, col = "blue", lwd = 2, lty = 2)
lines(x, loess_pred_1973, col = "darkgreen", lwd = 2)
legend("topright", legend = c("Data", "Kernel", "Loess"),
       col = c("blue", "blue", "darkgreen"),
       lty = c(1, 2, 1), lwd = c(1, 2, 2))
plot(x, y2003, type = "p", col = "red",
     xlab = "Week", ylab = "Max Temperature")
lines(x, ksmooth_2003$y, col = "red", lwd = 2, lty = 2)
lines(x, loess_pred_2003, col = "purple", lwd = 2)
legend("topright", legend = c("Data", "Kernel", "Loess"),
       col = c("red", "red", "purple"),
       lty = c(1, 2, 1), lwd = c(1, 2, 2))
```